Senior **Capstone Project**

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Abstract

The Internet, optimistically, should exist as a democratizing force in the world. However, as technology continues to occupy more and more of our lives, many older folks are left behind. This "digital divide," coined by Lloyd Morisset of the Markle Foundation, has impacted demographics without access to Internet technologies, including less tech-literate seniors. My aim in this year-long capstone project is to attempt to address this issue in local senior communities through my technology skills. My approach will be qualitative and personal; along with understanding the digital divide in a national sense, I intend to gain a deep understanding of local struggles with technology and develop a tool that will make the use of technology less daunting for older community members. Through this interdisciplinary project I will develop higher-level research skills, user-based design and career-specific skills, while strengthening my problem solving and communication to solve real world problems.

Rationale and Overview

These short paragraphs summarize my thoughts outlined in another document, that provides a more in-depth description of my approach, goals, and reasons I chose this project:

As a young person, I have a depth of experience with interacting with modern technology, having grown up with it and adapted to the nature of high-tech environments. However, despite my experience, interfacing with technology remains an issue. Even with the tools at my disposal and my enhanced ability to troubleshoot and research my tech issues, it remains a frustrating task. Now consider the same issues for a user that isn't able to draw the same conclusions that I can that I use to solve my tech problems; problems that seemingly come from nowhere, of which every action made on the screen submerges you deeper into the rabbit-hole of errors and jargon, unable to reset back to the familiar and trapped within cascading interfaces.

For seniors, interacting with technology and computers is daunting and at times embarrassing. In our shame-based and impatient culture, seniors are discouraged from engaging with new technologies, feeling inadequate or dumb for being unable to grasp modern devices. I've seen how tasks like setting up a new wireless printer, or transferring data from an old PC to a new one become herculean, and are often put aside or given-up on.

My goal is to understand and empathize with seniors who face struggles with technology and are unable to function fully in our increasingly technologically reliant society, and spend the school year on a capstone project to research, design, and build a device, application, or guideline that can support older adults in my community in their ventures with modern technology.

I am excited to see how this project will challenge me, and how it will allow me to learn more about electronics and user-centered design—skills that I hope to utilize in my career in electrical engineering—along with how my preconceived notions and prejudices when it comes to using technology will be revealed. I believe that my ethic of collaboration and understanding can help me in learning how to design a piece of technology for a group of individuals who often find disdain in using it, and ultimately rectify some of the fears and concerns that my local community members have when it comes to the developing technological landscape.

Essential Question

How can we use assistive technology to make technology less daunting for inexperienced older computer users?

Mentor / Outside Expert

Sam Tannous is a retired electrical engineer (Masters in DSP & Communications Systems), who has over thirty years of industry experience.

Demonstration of Learning

In order to document my learning process and developments following this capstone project, I intend to create and host a website. Through the website, there will be weekly reflections that detail progress in the project, in a "blog" style fashion. Additionally, I will include products of my learning where documents and media artefacts that serve to encapsulate a major aspect of the capstone project (e.g. Literature review abstract, survey questionnaire, images of I.T help).

Research Strategy

Essential Question: How can we use assistive technology to make technology less daunting for inexperienced older computer users?

Supplementary Research Questions:

- In what ways might adaptive technology be helpful in aiding older computer users struggling with technology?
- What is the biggest obstacle in using technology, computers, and the internet as an older person?
- What are some ways older people have adapted to technology?
- What frustrates older people most when using technology?
- In what ways do older people want or need to use technology and the internet?
- How can you address age-based issues with technology with assistive technology?

Research Plan

(refer to timeline)

My research plan encompasses three phases of information gathering. I will initially collect information from relevant literature about aging and technology. This will allow me to learn more about existing solutions, identified broad-spectrum issues with technology utilization by elders, and become familiarized with field-specific language. Next, I plan on making contact with local and government agencies and institutions that provide aid to seniors and aging people in Vermont, and gain more local knowledge of solutions and issues. The part of my research that will likely most inform my design choices will be my interactions with local communities like Valley Village in Townshend, VT, through interviewing staff and participants in the senior living space.

Research Design

Literature Review

My initial foray into research will be a review of the literature available of assistive technology tools and philosophies designed for seniors. The contents of my analysis will focus on the medical field and geriatrics, the consumer technology industry, and popular works of writing that connect to elders. This will allow me to gain a more global perspective, providing me with insight to common issues along with supplying direction for future qualitative research.

Focus

Due to the local nature of this capstone project, creating and maintaining an intentional community oriented approach to this project is important. Focusing on local issues and struggles that are unique to the local senior population in rural Southeastern Vermont both serves to amplify the possibility of community improvement through this capstone project and provides a naturally limited scope.

Limitations & Alternatives to Consider

Current limitations with the development of this capstone project lie in the current plan relying on Valley Village, and their community, who aren't obligated to reciprocate my goals with them.

Methodology

Contact

I plan on contacting projects and centers for aging care, and learning more about how they have addressed the issue of technology struggles. Through contacting these facilities, I hope to gain information surrounding existing solutions to elder tech struggles, and how I can help fill in the gaps using technology.

Local projects

Senior Solutions -

Senior Solutions is an organization that centers Southeastern Vermont seniors, and helps in addressing their struggles with life in our rural communities. I intend to make contact with the organization, and see how they have addressed the digital divide within senior demographics, and if they can provide me with helpful information that could serve a part in my research and development process. They will have a developed grasp of not only the struggles with technology where it comes to aging, but the specific struggles in the rural communities I live in, and the needs that need to be addressed within them.

Valley Village (formerly Valley Cares) -

Valley Village is a senior centered living community in Townshend, Vermont. My goal is to interact with the people living within this community, and learn from their experiences with technology. I plan on hosting I.T help sessions, and gathering qualitative research through that experience. If I can create a longer-term relationship with Valley Village through my research, I hope to involve them with the design process of my capstone project as well.

Bellows Falls Senior Center-

The Bellows Falls Senior Center is another senior living facility and community that falls under Senior Solutions. In the event that Valley Village is unable to meet my needs, I intend to contact them. While less local than Valley Village, the senior community in Bellows Falls is likely facing similar challenges with technology that other rural Vermont towns struggle with.

Government

Assistive Technology VT

The Vermont Assistive Technology program acts under the US Dept. of Health Services for Community Living, and helps in overcoming obstacles with technology that are based on disability or age-based needs. I intend to make contact with a representative of the

program in hopes to learn about their assistive technology solutions, underaddressed needs in our communities, and research regarding using technology to assist Vermont's aging population.

Interview & Survey

A significant part of my capstone project will involve understanding the issue of technology struggles through individual and group interviews with seniors and staff in institutions like Valley Village in Townshend, VT along with elder patrons and volunteers of the Moore Free Library (Newfane), as I have personal connections with many of them. I plan on holding IT help or Technology classes with Valley Village, and gather information through helping people. My goal is for this information to become the basis of my design phase, being able to utilize data to address specific and relevant issues with aging with technology.